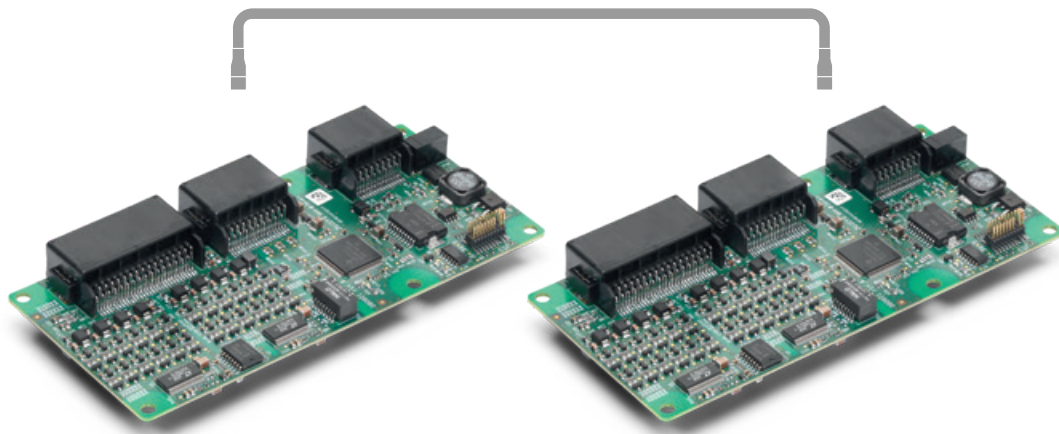




c-BMS with extended software functionality:

- Parallel pack capable up to 10 packs in parallel
- Battery swappable
- Heater control
- Hybrid SOC algorithm
- Advanced balancing, SOH, SOP, SOE algorithms

INTRODUCTION

The c-BMS24X combines the compact hardware and all the functionality of a traditional c-BMS with the latest features, software capability, and advanced Sensata Technologies | Lithium Balance proprietary algorithms developed for low Voltage BMS.

The c-BMS24X supports connecting up to 10 battery packs in parallel, providing flexibility in battery design along with improved safety and serviceability. The battery packs connected in parallel are hot swappable, eliminating downtime and range concerns normally associated with EVs due to the need of charging.

Hybrid SOC, along with advanced SOH, SOP, and SOE algorithms ensure that the c-BMS can maintain high measurement accuracy even with cell chemistries such as LFP, without the need of a full charge-discharge cycle or long rest time, essentially increasing overall system uptime.

Using the c-BMS24X Creator™ software, an advanced version of the traditional c-BMS' configuration software, the battery designer can define unique, application-specific battery settings, safety strategy, optimized battery performance and battery life. The c-BMS24X is cell agnostic both in terms of form factor and chemistry.

SAFETY

- Safety-rated key components
- Self-test and redundancy in safety critical measurement circuits
- Open circuit detection

BATTERY LIFE

- High frequency sampling of current at 10 mS allows optimal detection of pulses
- Powerful and intelligent dissipative balancing at 200 mA per cell
- Heater control

PERFORMANCE

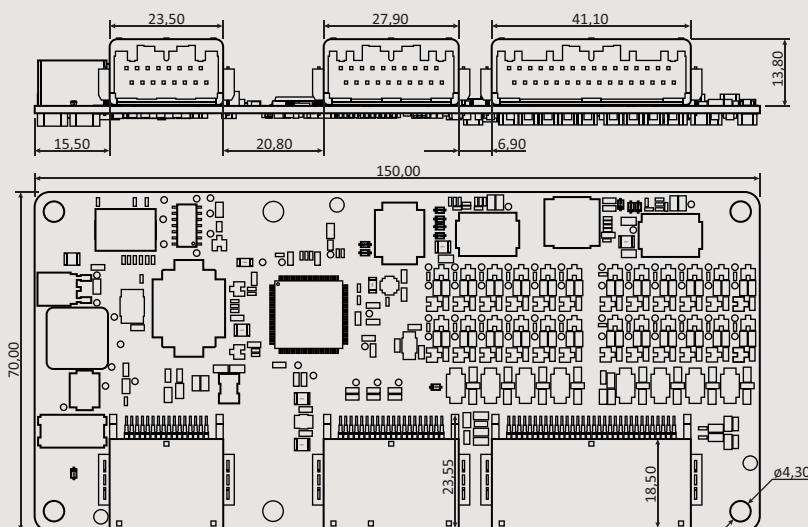
- Individual cell voltage measurement accuracy to within ± 1.6 mV at 25 °C
- Optimized low power consumption mode
- ± 1 °C accuracy in temperature measurement
- Hybrid SOC algorithm & OCV compensation with SOC accuracy reaching $\pm 0.5\%$
- Advanced SOH algorithm based on capacity fading and internal resistance estimation
- Advanced SOP and SOE algorithms
- Parallel pack support (up to 10 packs)
- Hot swap capability
- Advanced balancing algorithm allowing cells balancing at any time during the charge-discharge cycle

USABILITY

- RTC + logging of events, errors and warnings
- BMS Creator PC tool for easy configuration

Applications





c-BMS Compact Battery Management System for 24 cells

Dimensions in MM

PARAMETERS

Power supply
Range of high voltage measurement
Accuracy of high voltage measurement
Range of current measurement input Shunt
Accuracy of current measurement input Shunt
Range of current measurement input (Hall effect sensor)
Accuracy of current measurement input (Hall effect sensor)
Standby consumption (sleep mode)
Active consumption
Supported CAN communication type
Supported CAN speeds
CAN ports
External General Purpose I/O's
Charger control interfaces
Number of cells
Minimum detectable cell voltage
Maximum detectable cell voltage
Cell balancing topology
Cell balancing current
Cell voltage typical sampling time
Accuracy of single cell voltage
Range of Temperature measurements
Accuracy of cell temperature (NTC)
Patents
Temperature sensor channels
Dimension
Parallel pack support
Battery swap capability
Hybrid SOC algorithm

SPECIFICATIONS

6-35 V
0 - 120 VDC
±1 VDC
±200 mV
±0.5 mV -40 – 85 °C
0.0 – 5.0 V, 0.0 -2.5 V current in, 2.5 V – 5.0 V current out
±1.25 mV -40 – 85 °C
<2,5 mW
<2.7 W
CAN 2.0A/B 11 bit and 29 bit IDs
125, 250, 500, 1k kbit/sec
1
4 GP I/O (Active Low) and 4 inputs
CAN
Up to 24 Cells. Minimum 11 V
0 VDC
5 VDC
Dissipative
200 mA, at cell voltage 4.2 V
100 ms
±1,6 mV at 25 °C
-40 to +85 °C
±1 °C -40 – 85 °C
Granted: ZT 200780048774, EP 0781788.6, US 8.350.529
Up to 6
170 mm x 70 mm x 15 mm, 67 g
Yes
Yes
Yes

